Kshamaa Suresh

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kshamaas.github.io/Portfolio

PROFESSIONAL SUMMARY

I'm the kind of person who reads about a problem on a Monday and has something deployed by Thursday — not because I was asked to, but because building is how I think. Two years of production AI engineering at Ecolab (Claude API at scale, Databricks, Snowflake, 10M+ records daily) gave me real intuition for where AI systems break and what it actually takes to fix them in production. Now at Columbia finishing my MS in Data Science, I do published computational neuroscience research during the day and ship agentic AI systems at night — because the problems at the intersection of AI, healthcare, and cognition are too compelling to wait. I bring the rigor of someone who has seen systems fail in production and the drive of someone who builds anyway, even when the path isn't clear.

TECHNICAL SKILLS

Languages: Python, SQL, PySpark, R, Java, JavaScript, HTML/CSS

ML & AI: Machine Learning, Deep Learning, NLP, Generative AI, LLMs, Transformers, BERT, RAG, Prompt Engineering, Feature Engineering, Fine-Tuning (LoRA, BERT), Multi-Agent Systems, Agentic AI, Hallucination Mitigation, Guardrails, TensorFlow, PyTorch, Scikit-learn, HuggingFace, LangChain, ChromaDB, FAISS, BM25, XGBoost, SHAP

Data Engineering: ETL/ELT Pipelines, Data Modeling, Data Warehousing, Data Validation & Reconciliation, SAP-to-Snowflake Migration, SQL Server, QA Testing, Automated Regression Testing, Model Deployment

Cloud & Platforms: Snowflake, Databricks, Azure, AWS, Azure Repos, Snowflake Cortex, MLflow, Streamlit, Power BI, ThoughtSpot, HuggingFace Spaces, SQLite, Google Colab

Libraries & Tools: Pandas, NumPy, SciPy, Matplotlib, Seaborn, NetworkX, UMAP-learn, scanpy, muon/MuData, Groq API, Plotly, Git, REST APIs, Jupyter Notebook

DevOps & CI/CD: Azure Repos, YAML Pipelines, CI/CD Pipeline Configuration, Git Branching, Version Control, QA Testing, Test Automation

EDUCATION

M.S. Data Science | Columbia University*New York, NY · Sep 2025 – Dec 2026*

GPA: 3.50 · Applied ML, ML for Functional Genomics, Generative AI for LLMs, Exploratory Data Analysis, Algorithms for Data Science, Statistical Inference & Modeling, AI & Digital Transformations

B.Tech. Computer Science | MS Ramaiah University of Applied Sciences*Bangalore, India · Aug 2019 – Aug 2023*

GPA: 3.99/4.00 (Distinction, Top 5%) · Database Systems, AI, Deep Learning, Data Analytics, Python for Data Science, Data Structures & Algorithms

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | Barnard College – Columbia University*New York, NY · Jan 2026 – Present*

- Conduct statistical and graph ML analysis on Drosophila larval brain connectome (NetworkX, scikit-learn, NumPy, SciPy, Pandas); apply modularity maximization and stochastic block models for neural community detection across 1,000+ annotated neurons.
- Implement spatial analysis algorithms (k-nearest neighbor, Hopkins statistic, Ripley's K) to quantify synaptic clustering; co-authored abstract at CSHL Neuronal Circuits 2026 (p. 68); accepted poster at Cognitive Computational Neuroscience 2026; co-authoring peer-reviewed manuscript on network-level synapse localization.
- Consolidated 10+ fragmented research scripts into 3 reusable, documented Python modules under PI Dr. Gabrielle Gutierrez, reducing analysis runtime by ~40% and ensuring full reproducibility across experiments.

Associate Data Engineer – Business Process Analytics | Ecolab*Bangalore, India · Jan 2025 – Jul 2025*

- Engineered production Generative AI data validation system using Anthropic Claude API on Databricks and Snowflake Cortex AI (Mistral-large2), achieving 70% improvement in anomaly detection accuracy across 10M+ records.
- Built end-to-end NLP pipeline to parse and persist LLM outputs into Snowflake tables, automating ThoughtSpot BI dashboards for 50+ stakeholders; compressed analytics cycle from days to 3 hours. Managed end-to-end model deployment and production rollout of Streamlit dashboards on Azure for self-service visualization.
- Engineered Python pipeline monitoring system parsing Snowflake execution logs to fire real-time email alerts in <120s, reducing downtime 40%+. Built ML classification framework correcting erroneous MDM records across 5M+ rows, cutting manual review 80%.

Associate Data Engineer | Ecolab*Bangalore, India · Jul 2023 – Dec 2024*

- Designed and executed end-to-end QA testing strategy for SAP-to-Snowflake ETL pipelines, performing systematic dev-to-production data validation across all transformation layers; achieved 98% data accuracy via primary key join reconciliation.
- Built SQL Server-to-Snowflake data reconciliation system performing primary key join comparisons across millions of records; auto-generated discrepancy reports compressed migration timeline by 60%.
- Engineered Python-based automated data quality framework detecting schema drift, transformation errors, and statistical anomalies across 1,000+ Snowflake views; reduced QA cycle time by 90% and cut manual testing effort from days to hours.
- Configured and maintained CI/CD pipelines using Azure Repos and YAML across 3 environments (dev, staging, production), reducing manual deployment steps by 70% and cutting release cycle time from days to under 2 hours.

Data Engineering Intern | Ecolab*Bangalore, India · Jan 2023 – Apr 2023*

- Built Python regression automation tool certifying data transformations across Snowflake layers; selected as final university thesis (50% of semester grade).

PROJECTS

ClaimSense – Agentic Healthcare Payment Integrity System*huggingface.co/spaces/kshamaasuresh/ClaimSense*

- Problem: CMS reported $31B in improper Medicare payments in FY2023. Existing tools flag anomalies but stop there — reviewers manually reason through why a claim is suspicious, creating a costly bottleneck between detection and decision.
- Architecture: Three-stage pipeline — rule-based detection across 6 anomaly categories (diagnosis-procedure mismatch, duplicate billing, unbundling, amount outliers, future dates, high-frequency billing) → Groq LLM (llama3-70b) chain-of-reasoning explanation per flagged claim → RAGAS-style faithfulness evaluation rejecting responses that reference facts not present in the input. Chose rule-based detection over ML classifier to keep the system auditable in a clinical context.
- What broke: Early LLM prompts generated plausible but unfaithful explanations, inventing billing history not in the claim record. Fixed with strict grounding instructions and a claim-level faithfulness gate.
- Metric: Flags claims across 6 anomaly types with confidence scores and risk levels (Low/Medium/High/Critical); LLM-generated reasoning and recommended reviewer action per flagged claim; 4-page interactive app deployed live on HuggingFace Spaces free tier. Stack: Python, Groq API, LLaMA3-70b, Streamlit, Pandas, RAGAS.

CognitivePulse – Preventive Brain Health ML & RAG Pipeline*huggingface.co/spaces/kshamaasuresh/CognitivePulse*

- Problem: Brain health interventions are most effective years before symptom onset, but existing tools offer generic lifestyle advice untethered from individual biomarker profiles — no personalization, no explainability, no faithfulness guarantees on generated health claims.
- Architecture: Five-stage pipeline — biomarker intake → XGBoost risk stratification with SHAP TreeExplainer waterfall charts (trained on 2,149-patient Alzheimer's dataset, 33 features, 5-fold stratified CV) → modifiability-weighted intervention priority ranking → RAG coaching brief using e5-large-v2 embeddings, FAISS retrieval over 14-paper dementia prevention corpus, and Groq LLM inference → patient-specific Q&A chatbot with RAGAS-style claim-level faithfulness evaluation (SUPPORTED/PARTIAL/UNSUPPORTED) before display.
- What broke: SHAP global importance charts mislead on individual patients — a feature important globally may not drive a specific patient's risk score. Fixed by switching to per-patient waterfall charts using TreeExplainer.
- Metric: Full five-stage inference pipeline runs at zero cost on HuggingFace Spaces free CPU tier with synthetic data fallback; every health claim faithfulness-checked before rendering. Stack: Python, XGBoost, SHAP, FAISS, Groq API, Streamlit, Plotly, HuggingFace.

SlideScholar – Multimodal RAG Study Assistant*huggingface.co/spaces/kshamaasuresh/SlideScholar | github.com/KshamaaS/SlideScholar*

- Problem: Generic AI tools hallucinate answers on course-specific questions, scoring only 0.35 faithfulness against grounded QA pairs from actual lecture slides.
- Architecture: Two-stage multimodal ingestion — LLaVA-7B vision descriptions + pdfplumber text extraction merged per slide — before embedding with intfloat/e5-large-v2. Pure text extraction silently discards diagrams and equations that carry the majority of information in STEM slides.
- What broke: Mistral-7B flashcard generation collapsed to identical cards across topics when JSON prompt lacked structural constraints — fixed by ending the prompt with a bare [token to force array generation, plus a regex fallback parser for truncated outputs.
- Metric: Evaluated on 50 grounded QA pairs from actual course slides — hit rate 0.80, retrieval F1 0.75, answer relevance 0.78; 43-point faithfulness improvement over ChatGPT baseline (0.72 vs 0.35). Entire inference stack runs at zero cost. Stack: Python, LLaVA-7B, Mistral-7B, HuggingFace Inference API, FAISS, Gradio, PyTorch.

Dual-UMAP Multimodal Genomics Integration*github.com/KshamaaS/Multiome_UMAP_Project*

- Novel UMAP extension with Coupling Regularizer on 34,774-cell single-cell dataset (19,303 RNA genes + 344,592 ATAC peaks); 58% Silhouette Score improvement over WNN baselines. Sparse pipeline reduced RAM 98.8% (96 GB → 1.2 GB). Stack: Python, scanpy, UMAP-learn, NumPy, SciPy.

Tiny Transformer – Shakespeare Text Generation*github.com/KshamaaS/TinyTransformer*

- Transformer (3 layers, 8 attention heads, 722K params) with WordPiece tokenization; validation perplexity 45.25 (11× better than random). GPU acceleration, cosine annealing LR schedule. Stack: PyTorch.

PUBLICATIONS

- Gutierrez G.J., San Francisco P., Suresh K. "Relating Network-level Organization to Synapse Localization" Cold Spring Harbor Laboratory – Neuronal Circuits 2026 Abstract Book, p. 68. (2026)
- Suresh K. "A Memo of Memories" Self-published anthology of poems and short stories. Amazon KDP (ISBN: 979-8316935727). (2025)
- Suresh K. et al. "Smart Irrigation System using Soil Moisture Sensor" IJERCSE Vol.10, Issue 7. (2023)